

Kevin Liu Rodrigues

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I am a scientist with a PhD in computational and statistical physics and solid quantitative basis. I first started programming in Tcl/Tk during my masters degree for interfacing with various electronic equipments and soon moved to Python, which I have been using ever since.

Inspired by advances in artificial intelligence in the form of AlphaGo, in my PhD I moved to statistical and computational physics, implementing various dynamical systems simulations in C, leading me to a collaboration with the University of Michigan, where I was able to give a mathematical description for some systems of oscillators which corroborated simulation results.

This led me to a diverse experience in data science, machine learning and software development: In 2018 I joined Olivia AI to develop sentiment analysis and a chatbot before the advent of LLM's. By 2020 I was hired at Rappi to develop life time value models and user metrics to aid in the company's expansion in South America, and finally in 2021 I joined Spacetime Labs (now Moray AI) to work with machine learning for precision agriculture and crop management through climate and sensor data.

EXPERIENCE

Moray (2021 – Present)

Data Scientist and Engineer

- Machine learning and predictive models for precision agriculture and crop management.
- Data pipelines for gathering and processing spatio-temporal data from sources such as NASA, Copernicus, ECMWF, etc.
- Time series analysis and forecast.
- Machine learning pipelines from training to deployment using mainly MLflow and AWS.

Rappi (2020 – 2021)

Data Scientist

- Development of user life-time value models and user metrics.
- Large dataset processing and visualizations including spatio-temporal maps and dashboards.

Olivia AI (2018 – 2018)

Data Scientist

- Development of a chatbot in the very early stages of AI, experimenting with sentiment analysis, string distance metrics and word embeddings.
- Classification models for text, including Neural Networks, SVM's and ensemble methods.

Federal University of Minas Gerais (2015 – 2016)

Researcher and Engineer

- Development of a high power near-field microscope for Hewlett-Packard (HP) under coordination of professor Gilberto Medeiros

University of Campinas (2010 – 2012)

Scientific initiation

- Development of pico Faraday detection electronics for capacitive sensing.
- Development of electrostatic lensing for ion beam sampling and focusing.

TRIU (2012 – 2014)

Volunteer math teacher

- Math classes for low income students aiming to join top universities.

EDUCATION

Doctorate in statistical physics | UFMG and UMICH (2016 – 2021)

(Advisors: Ronald Dickman and Kevin Wood)

Master of science in applied physics | University of Campinas (2012 – 2015)

Bachelor of science in physics | University of Campinas (2008 – 2012)

SKILLS

Programming languages: Python, SQL, C, C++, TCL/TK, JavaScript

Tools: MLflow, Docker, Amazon Web Services (AWS), Airflow, Kubernettes, Parametric Modelling (CAD)

Languages: Portuguese (native), English (fluent spoken/written)